

Accuracy of a requirements ambiguity detector: case of a pet social network

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Abstract

Context: Ambiguity is one of the most persistent defects in natural-language requirements specifications, and rule-based detectors are frequently proposed as a low-cost mechanism to flag it before requirements reach design. Evidence on how closely such detectors reproduce human expert judgement remains scarce for Spanish-language specifications produced in Latin American settings.

Objective: This study measures the extent to which a rule-based ambiguity detector replicates the consensus judgement of independent human experts over the complete requirements corpus of MundiPets, a real social network for responsible pet adoption and breeding, and characterises the patterns of disagreement between the two.

Methods: We conducted a paired quasi-experiment over 61 requirements (27 functional, 16 non-functional, 9 design constraints and 9 legal requirements) specified according to an eight-attribute template. A purpose-built rule-based detector, implemented in Python and frozen before data collection, classified each requirement as ambiguous or non-ambiguous using four lexical and syntactic rules. In parallel, three independent software engineers, blind to the detector output and to one another, classified the same requirements in randomised order

using a shared rubric. The protocol was preregistered on the Open Science Framework before any data were collected. Agreement was quantified with precision, recall, F_1 , Cohen’s κ and Fleiss’ κ .

Results: The detector flagged 32.79% of requirements as ambiguous (20/61) against 26.23% (16/61) for the expert consensus, yielding accuracy = 0.8033 (95% CI [0.7049, 0.9016]), precision = 0.600, recall = 0.750 and $F_1 = 0.667$ for the ambiguous class. Agreement between detector and consensus was moderate (Cohen’s $\kappa = 0.530$, 95% CI [0.280, 0.737]) and a chi-squared test of independence found the detector significantly associated with expert consensus beyond chance ($\chi^2 = 15.037$, d.f. = 1, $p = 0.0001$, $\phi = 0.497$). The expert panel itself agreed almost perfectly (Fleiss’ $\kappa = 0.857$), which strengthens confidence in the reference standard the detector is compared against. Coincidence with expert consensus was strongest for legal requirements (9/9, 100%) and non-functional requirements (14/16, 87.5%), and weakest for design constraints (5/9, 55.6%).

Conclusion: A four-rule lexical-syntactic detector achieves moderate, statistically significant agreement with an almost-perfectly agreeing expert panel on a real, Spanish-language requirements corpus, performing best on non-functional and legal requirements and worst on design constraints, where ambiguity is more often a matter of missing procedural detail than of detectable lexical patterns. The result supports using such a detector as a first-pass triage tool in a review workflow, while showing that design constraints in particular still require human judgement.

Keywords: Requirements Engineering, Empirical Software Engineering, Natural Language Processing, Interrater Agreement, Requirements Quality

1 Introduction

MundiPets is a social network for responsible pet adoption and breeding, currently under specification for veterinary practices and pet owners in the province of Los Ríos, Ecuador. The system lets owners publish verified pet profiles, lets registered veterinary practices validate the associated medical records, and supports adoption and breeding requests through a staged workflow that includes compatibility assessment by an artificial-intelligence component. Requirements engineering matters acutely in this domain because a defect in a requirement about medical-record validation or about the visibility of an owner’s location does not merely degrade usability: it propagates into a system that handles animal health data and personal data protected under Ecuador’s Organic Law on Personal Data Protection [?]. Specifying such a system with verifiable, unambiguous requirements is therefore a precondition for lawful operation rather than a stylistic preference.

Ambiguity remains one of the defects most consistently reported in natural-language requirements. The systematic mapping study of Montgomery et al. [?] shows that, although requirements quality has been studied extensively, empirical work concentrates on a small number of quality attributes and rarely validates automated detection against human judgement on complete, real specifications. Atoum et al. [?]

reach a convergent conclusion in their systematic review of requirements quality assurance, identifying the scarcity of validated automated support as an open challenge. Ferrari et al. [?] demonstrated that ambiguity in elicitation is entangled with tacit knowledge, which suggests that purely lexical detection has an intrinsic ceiling; yet the size of that ceiling has not been quantified for specifications written in Spanish, nor for corpora produced by novice analysts, the setting in which a large share of requirements documents are in fact written. This is the gap the present study addresses.

This paper makes three contributions, each grounded in evidence collected by the authors and released as an open dataset:

1. a quantified comparison between a rule-based ambiguity detector and the blind consensus of three independent human experts over a complete requirements corpus of 61 items — functional requirements, non-functional requirements, design constraints and legal requirements — written in Spanish;
2. a characterisation of the disagreement patterns between automated and human classification, reported per requirement and per rule, which exposes the specific linguistic phenomena that a rule-based approach systematically over- or under-flags;
3. a replication package — corpus, detector output, expert classifications, analysis scripts and preregistered protocol — released under CC BY 4.0 so that the study can be repeated on other corpora and other languages.

The study answers two research questions:

- RQ1** With what precision, recall and F_1 does a rule-based automatic detector replicate the consensus judgement of human experts when classifying the requirements of MundiPets as ambiguous?
- RQ2** Which disagreement patterns predominate between the automatic detector and expert judgement, and what do they reveal about the limits of a rule-based approach relative to human judgement in identifying ambiguity?

The remainder of the paper is organised as follows. Section ?? positions the study within the closest related work. Section ?? describes the experimental design, participants, instruments and analysis plan. Section ?? reports the results, Section ?? interprets them, Section ?? discusses threats to validity, and Section ?? concludes.